L1 SOC ALERT TRIAGE REPORT

Ubuntu Authentication Failures Analysis in Splunk Enterprise

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1. EXECUTIVE SUMMARY & OBJECTIVE

As part of my DG Interns Hub Week 5 practical assessment, I conducted a hands-on investigation as a Tier 1 (L1) SOC Analyst. The objective was to set up a SIEM monitoring environment, configure Linux authentication log ingestion, query authentication failures using Splunk Search Processing Language (SPL), analyze process-level events, and make a formal alert triage determination.

> **Investigation Summary**
> During log analysis of Linux authentication records (`/var/log/auth.log`), I identified **5 total authentication failure events** associated with the user account `gunjan` on host `ubuntu-soc`. Analysis revealed 4 failures generated by the graphical display manager (`gdm-password`) and 1 failure generated by privilege elevation (`sudo`).

2. LAB ENVIRONMENT & SYSTEM CONFIGURATION

Component	Technical Specification / Configuration Details
SIEM Platform	Splunk Enterprise 10.6.1 (Standalone Linux Deployment)
Operating System	Ubuntu Linux running inside Oracle VirtualBox VM
Hostname & IP	`ubuntu-soc` (Access via `http://localhost:8000`)
Target Log Source	`/var/log/auth.log` (File Size: ~51 KB)
Splunk Index & Type	Index: `main` | Sourcetype: `linux_secure`
Monitored User	`gunjan` (UID: 1000)

3. PRACTICAL IMPLEMENTATION & STEP-BY-STEP SETUP

Step 1: Splunk Installation & Service Verification

I installed Splunk Enterprise on my Ubuntu virtual machine, configured the admin user, and resolved permissions. To follow Linux security best practices, I created a dedicated `splunk` system user and added it to the system log monitoring group (`adm`) using:

```
sudo usermod -aG adm splunk
```

I verified that the Splunk service daemon (`splunkd`) was running correctly and checked memory consumption using terminal commands.

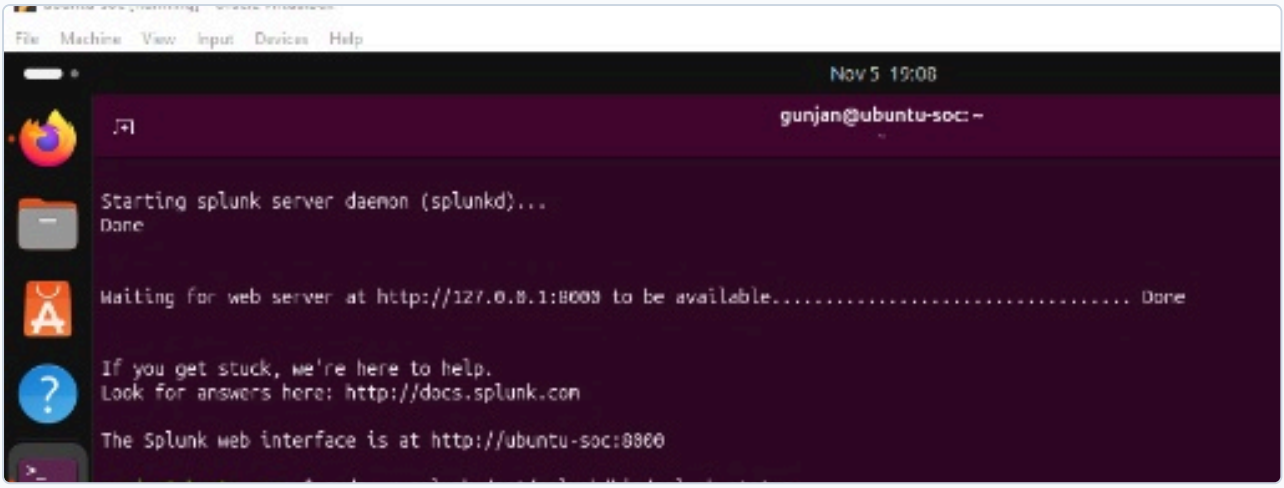


Figure 1: Terminal verification of Splunk daemon status, permissions, and log file availability.

Demonstrates `sudo -u splunk ... splunk status` confirming splunkd is running (PID 13902), memory usage via `free -h`, and log file sizes (`/var/log/auth.log` at 51KB and `/var/log/syslog` at 3.2MB).

Step 2: Understanding Splunk Internal Events

Before ingesting system security logs, I verified that Splunk was correctly indexing its internal telemetry by searching the internal index:

```
index="_internal"
```

This initial search returned **39,392 matching events** across internal logs. Next, I executed an SPL query to break down these events by log type:

```
index="_internal" | stats count by sourcetype | sort -count
```

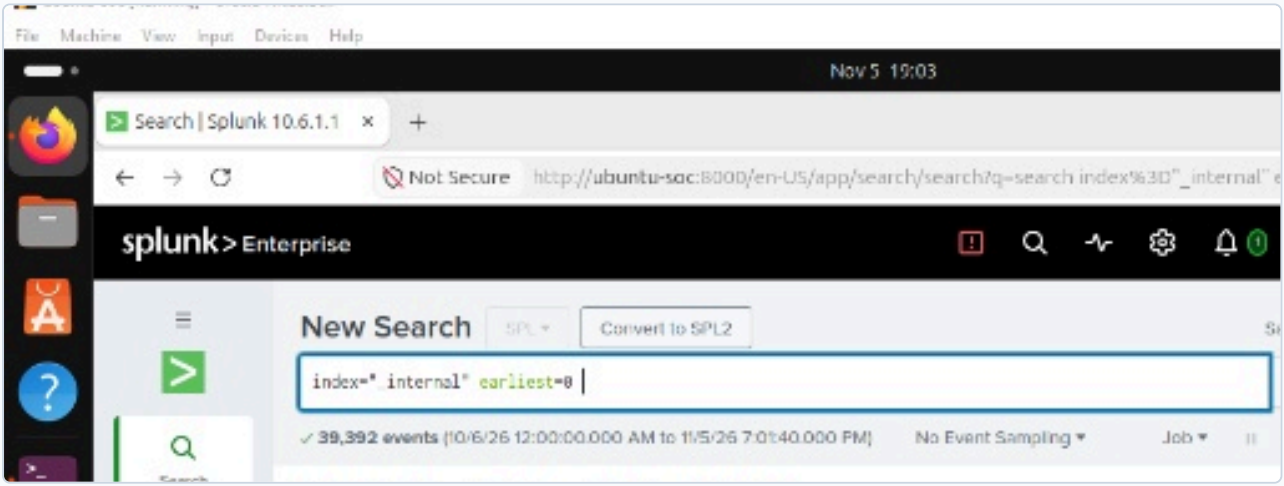


Figure 2: Splunk Web search verifying internal index event collection (39,392 events).

Validates that Splunk Enterprise is actively processing data within the `index="_internal"` default store.

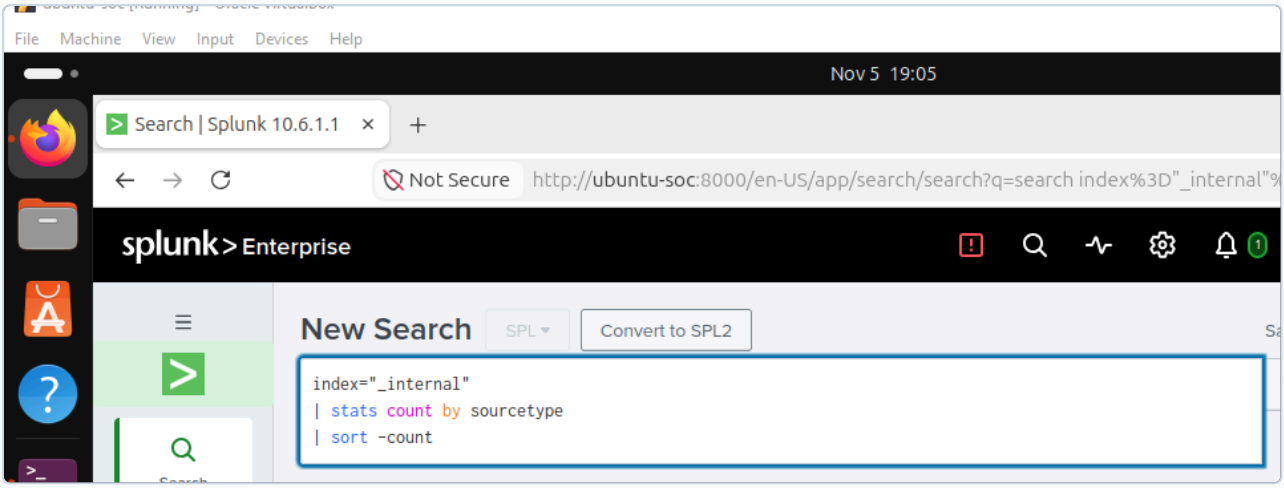


Figure 3: SPL search breaking down internal events by sourcetype (`stats count by sourcetype`).

Results show `splunkd` (25,833 events), `node:supervisor` (4,925 events), `node:sidecar:kvstore_pdl:stdout` (3,630 events), and `splunkd_access` (2,209 events).

Step 3: Linux Authentication Log Ingestion

I configured Splunk to monitor Ubuntu authentication logs via **Settings → Add Data → Monitor → Files & Directories**. I selected `/var/log/auth.log`, set monitoring to **Continuously Monitor**, assigned the sourcetype as `linux_secure`, host as `ubuntu-soc`, and directed events to index `main`.

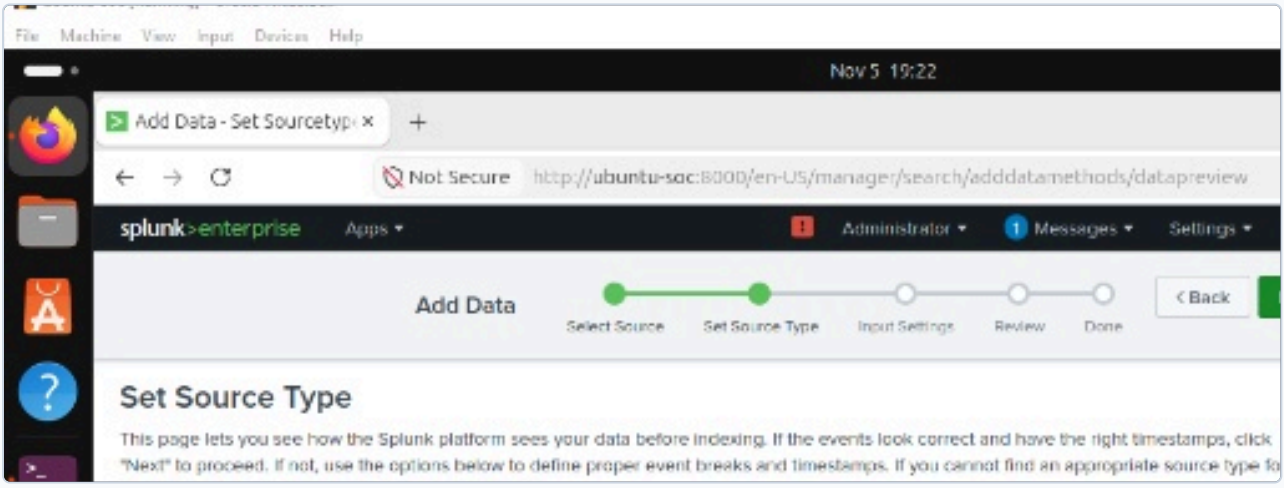


Figure 4: Data preview window during `/var/log/auth.log` ingestion configuration.

Shows real-time parsing of Linux PAM authentication events, timestamp extractions, and session events.

Step 4: Verification of Data Ingestion

To confirm that authentication logs were successfully indexed in the `main` index under `linux_secure`, I ran:

```
index=main sourcetype=linux_secure | head 20
```

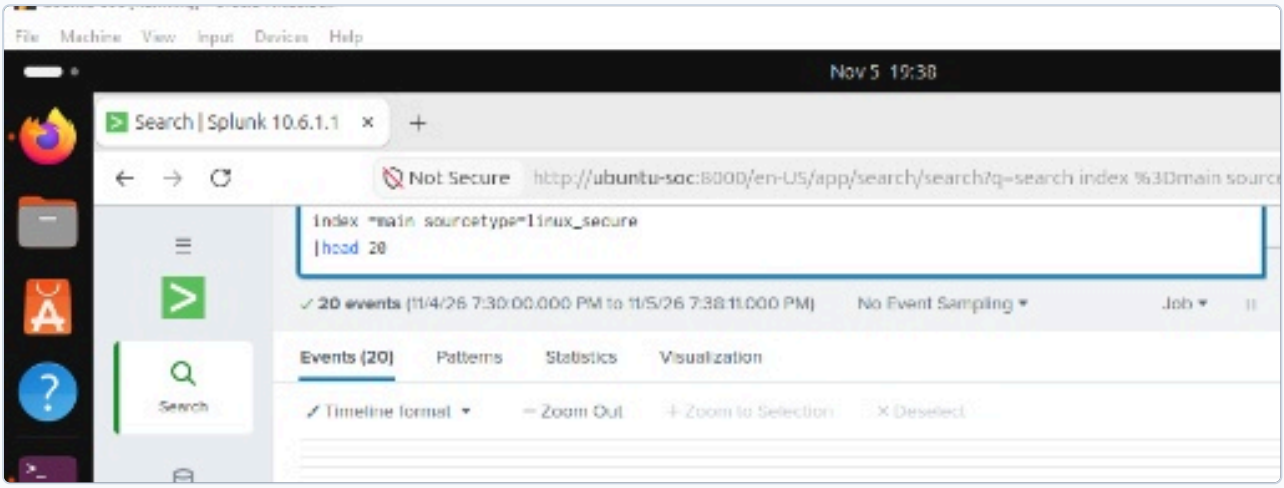


Figure 5: Verification of ingested authentication events in Splunk Web (`head 20`).

Confirms live log stream ingestion from `/var/log/auth.log` showing `pam_unix` session events for user `gunjan` and `root`.

4. DETECTING & ANALYZING AUTHENTICATION FAILURES

To isolate failed login attempts, I executed a targeted SPL filter search:

```
index=main sourcetype=linux_secure ("Failed password" OR "authentication failure")
```

This query matched exactly **5 security events** in the index.

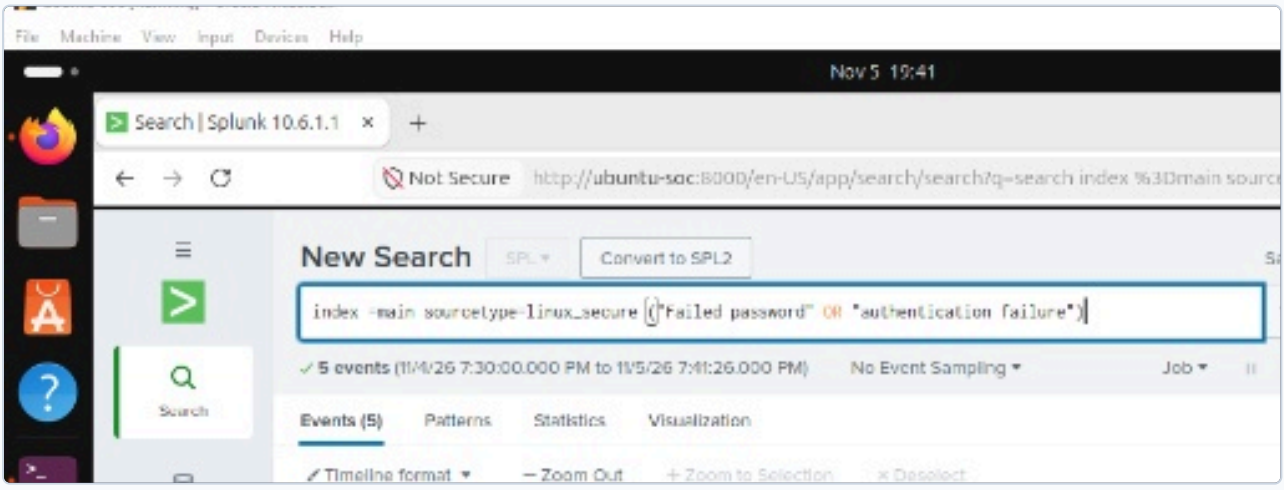


Figure 6: Search results isolating 5 authentication failure events.

Shows failure entries recorded by `sudo: pam_unix(sudo:auth): authentication failure` and `gdm-password: pam_unix(gdm-password:auth): authentication failure`.

Step 6: Statistical Analysis by Process & Username

To understand which authentication mechanisms generated these failures, I executed a statistical aggregation query:

```
index=main sourcetype=linux_secure ("Failed password" OR "authentication failure") | stats
count by process, logname | sort -count
```

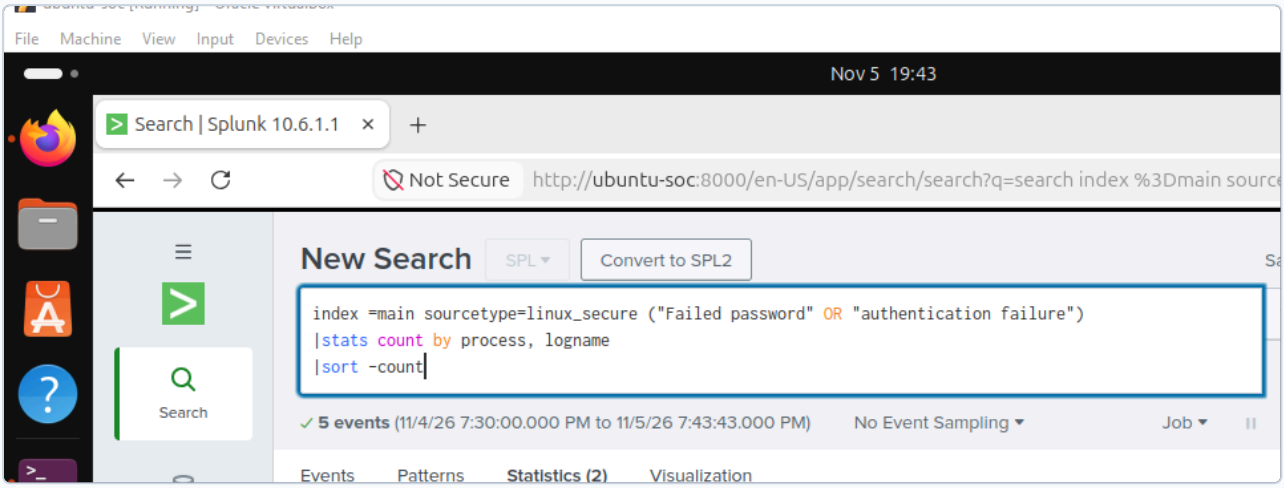


Figure 7: Aggregated failure count table by process and username (`stats count by process, logname`).

Statistical output: `gdm-password` (4 events) and `sudo` (1 event), both for user `gunjan`.

5. TECHNICAL CONCEPTS & AUTHENTICATION ANALYSIS

PAM (Pluggable Authentication Modules): A flexible Linux framework that handles user authentication. The module `pam_unix` handles standard Unix password checks. **gdm-password:** GNOME Display Manager's PAM service. Failures under `gdm-password` represent incorrect password entries at the local graphical lock screen. **sudo:** Command-line tool allowing users to execute commands with elevated privileges. Failures under `sudo` represent incorrect password attempts in the terminal.

6. MITRE ATT&CK MAPPING

Tactic	Technique ID & Name	Relevance & Assessment Note
Credential Access	**T1110 – Brute Force**	Potential mapping for investigation. The presence of 5 failed authentication logs reflects credential access verification efforts, but low volume indicates non-automated activity.

7. L1 SOC ALERT TRIAGE DETERMINATION

Affected User & Host:	User: `gunjan` | Host: `ubuntu-soc`
Log Source & Index:	`/var/log/auth.log` | Index: `main` (Sourcetype: `linux_secure`)
Observed Breakdown:	Total Failures: 5 (4× `gdm-password`, 1× `sudo`)
Severity Rating:	LOW (Needs Further Assessment)
Triage Classification:	**Needs Further Investigation – Brute Force Not Confirmed**
Analyst Assessment:	The 5 failed authentication events occurred locally on desktop interfaces (`gdm-password`) and terminal prompts (`sudo`). There is zero evidence of an external remote IP, automated script execution, or network brute-force activity. The events represent benign user password typos during local system usage.

8. RECOMMENDED ACTIONS & LEARNING OUTCOMES

L1 Recommended Response Actions:

Continue baseline monitoring of `/var/log/auth.log` in Splunk to detect any sudden volume spikes. Monitor for subsequent successful logins (Event `session opened for user gunjan`) following failure clusters. If SSH (`sshd`) failures from external IP addresses appear in future logs, initiate IP reputation lookups on AbuseIPDB and escalate to Tier 2.

Student Learning Outcomes:

Through this practical task, I gained hands-on experience installing Splunk Enterprise on Ubuntu Linux, managing service permissions, configuring continuous file ingestion for `/var/log/auth.log`, writing custom SPL queries, analyzing Linux PAM authentication logs, and applying objective triage principles without jumping to false conclusions.

References: Splunk Enterprise Documentation, Ubuntu System Administration Guide, MITRE ATT&CK Framework (T1110).